

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	B	A	A	A	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,D	C,D	A,C,D	A,C,D	B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	01260.00	25.00	1800.00	360.00	0.88 to 0.89	0.00

PART-2 : CHEMISTRY

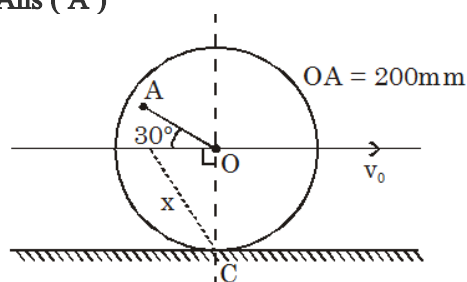
SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	A	C	A	B	D	C
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,C	A,C,D	A,B,C	B,C	C	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	6.00	4.00	5.00	1.00	2.00

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	B	D	D	C	C	A
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,C	B,C,D	A,D	A,C	A,B,D	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	6.00	4.00	6.00	6.00	21.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I(i)

3. Ans (A)



$$\begin{aligned}
 x^2 &= R^2 + OA^2 - 2ROA \cos 120^\circ \\
 &= 300^2 + 200^2 - 2 \times 300 \times 200 \times \frac{1}{2} \\
 &= 19 \times 10^4 \\
 x &= 100\sqrt{19} \\
 v &= \omega x \\
 v_0 &= \omega R \\
 \Rightarrow v &= \frac{v_0 x}{R} = 3 \times \frac{100\sqrt{19}}{300} = \sqrt{19} \text{ m/s}
 \end{aligned}$$

4. Ans (A)

$$C_1 V_1 + C_2 V_2 = (C_1 + C_2) V$$

$$\frac{C_1}{C_2} V_1 + V_2 = \left(\frac{C_1}{C_2} + 1 \right) V$$

$$x \times 300 + 100 = (x + 1) \times 250$$

$$50x = 150$$

$$x = 3$$

5. Ans (A)

$$\frac{v}{R_1} = i_1 \quad \frac{v}{R_2} = i_2 \quad \frac{v}{R_3} = i_3$$

$$\frac{\mu_0 i_1 i_2}{2\pi x_1} = \frac{\mu_0 i_2 i_3}{2\pi x_2}$$

$$\frac{x_1}{x_2} = \frac{i_1}{i_3} = \frac{R_3}{R_1} = \frac{5}{3}$$

6. Ans (B)

$$R_{eq} = \frac{2 \times 3}{2+3} + 12.8 = 4\Omega$$

$$i_{battery} = \frac{6}{4} = 1.5 \text{ A}$$

$$i_2 = \frac{3}{2+3} i_{battery} = \frac{3}{5} \times 1.5 = 0.9 \text{ A}$$

PART-1 : PHYSICS

SECTION-I(ii)

7. Ans (B,D)

$$\tan \theta = \frac{B_{coil}}{B_H} = \frac{\frac{\mu_0 N i}{2R}}{B_H}$$

$$= \frac{4\pi \times 10^{-7} \times 100 \times i}{2 \times \pi \times 5 \times 10^{-5}} = 0.4i$$

8. Ans (C,D)



$$B_1 = \frac{\mu_0 J R}{2}$$

$$J = nev = \rho \omega R$$

$$B_1 = \frac{\mu_0 \rho \omega R^2}{2}$$

Rotation $\Rightarrow$ many solenoids

$$dB = \mu_0 \frac{dI}{\ell} = \mu_0 \frac{dq \times \omega}{2\pi \ell} = \frac{\mu_0 \rho \times 2\pi x dx \omega \ell}{2\pi \ell}$$

$$B_0 = \frac{\mu_0 \rho \omega R^2}{2}$$

9. Ans (A,C,D)

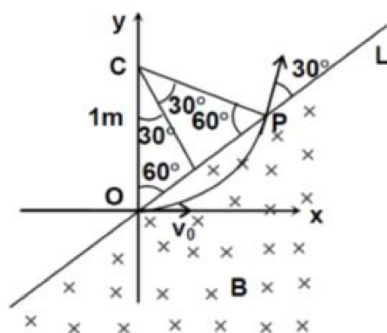
$$OB = 1 \text{ m}$$

and time period

$$T = \frac{2\pi m}{qB} = 2\pi \text{ sec}$$

Time spent by the particle in the magnetic field is

$$t = \frac{\pi}{3\omega} = \frac{\pi m}{3qB} = \frac{\pi}{3} \text{ sec}$$



10. Ans (A,C,D)

L about point of contact is conserved

$$L_i = mv_0 R - \frac{2}{5} m R^2 \omega_0$$

if $L_i > 0 \Rightarrow$ it will continue to roll forward.

$L_i = 0 \Rightarrow$ stops $L_i < 0 \Rightarrow$ Comes back

$$\Rightarrow mv_0 R < \frac{2}{5} m R^2 \omega_0$$

$$\frac{2}{5} m R^2 \alpha = \mu mg R$$

$$\alpha = \frac{5\mu g}{2R}$$

$$0 = \omega_0 - \alpha t$$

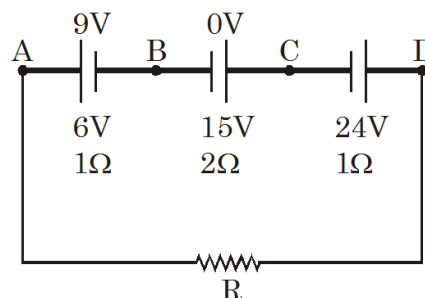
$$a_C = -\mu g$$

$$0 = v_0 - \mu g t$$

$$t = \frac{v_0}{\mu g}$$

$$\text{so } t = \frac{2R\omega_0}{5\mu g}$$

11. Ans (B,C)



$$9 = 6 + i \times 1$$

$$\Rightarrow i = 3 \text{ A}$$

$$\Rightarrow 33 = 3 \times (4 + h)$$

$$\Rightarrow R = 7\Omega$$

$$V_{BD} = 39 - 3i = 30 \text{ V}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (01260.00)**

Potential difference without introducing the slab is given by

$$V_0 = E \times d = 300 \times 5 = 1500 \text{ volt}$$

The capacitance C_0 is given by

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 A}{0.05} \text{ farad}$$

$$\therefore \epsilon_0 A = 0.05 C_0 \quad \dots (1)$$

when the slab is inserted, the capacitance C is given by

$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{k}} \text{ where } t = \text{thickness of slab}$$

$$= \frac{0.05 C_0}{0.04 + \frac{0.01}{5}} = \frac{25 C_0}{21} \quad \dots (2)$$

$$\text{Now } C_0 V_0 = CV$$

$$C_0 \times 1500 = \frac{25 C_0}{21} \times V$$

$$\text{or } V = \frac{1500 \times 21}{25} = 1260 \text{ volt}$$

2. **Ans (25.00)**

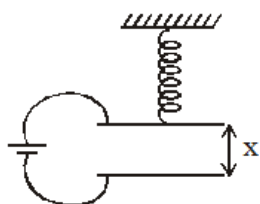
$$i_1 = i \times \frac{R_2}{R_1 + R_2} = 3 \times \frac{30}{30 + 60} = 1 \text{ A}$$

$$i_1^2 R_1 t = ms \Delta T$$

$$1^2 \times 60 \times 7 \times 60 = 0.240 \times 4200 \Delta T$$

$$\Delta T = \frac{600}{24} = 25^\circ \text{C}$$

3. **Ans (1800.00)**



$$\int \frac{q^2}{2A\epsilon_0} ds - \frac{1}{2} k \Delta \ell^2 = 0$$

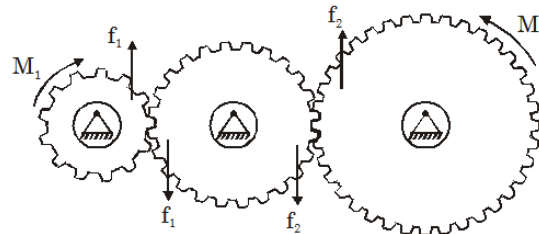
$$-\frac{\epsilon_0 A}{2} V^2 \int_d^{d/2} \frac{dx}{x^2} = \frac{1}{2} k \Delta \ell^2$$

$$-\frac{\epsilon_0 A V^2}{2} \left[-\frac{1}{x} \right]_d^{d/2} = \frac{1}{2} k \left(\frac{d}{2} \right)^2$$

$$\frac{\epsilon_0 A V^2}{2} \left[\frac{1}{d} \right] = \frac{1}{2} k \frac{d^2}{4}$$

$$v = \sqrt{\frac{k d^3}{4 \epsilon_0 A}} = \sqrt{\frac{36 \times 10^{-9}}{4 \times 8.85 \times 10^{-12} \times \pi \times 10^{-4}}} \\ = \sqrt{36 \times 10^{-9} \times 9 \times 10^9 \times 10^4} = 1800 \text{ V}$$

4. **Ans (360.00)**

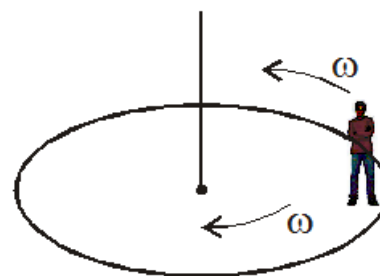


$$f_1 = f_2$$

$$\Rightarrow M_1 = f_1 r_1$$

$$M_3 = f_2 r_3 = f_1 r_3 = M_1 \frac{r_3}{r_1} = 60 \times \frac{3}{0.5} = 360 \text{ Nm}$$

5. **Ans (0.88 to 0.89)**



$$L_i = 0 = mvR - I\omega$$

$$v + R\omega = 2$$

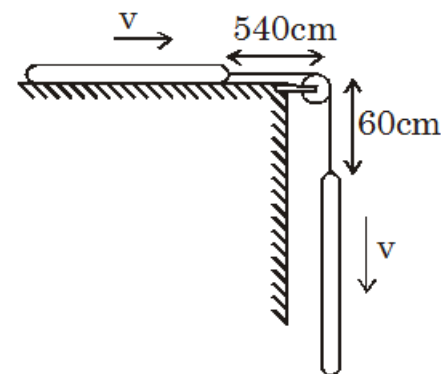
$$v = 2 - R\omega$$

$$0 = 80 \times 2 (2 - R\omega) - \frac{1}{2} \times 20 \times 2\omega$$

$$0 = 320 - 360 \omega$$

$$\omega = \frac{8}{9} \text{ rad/s} = 0.88$$

6. **Ans (0.00)**



v is || to length

$$\Rightarrow \epsilon = 0$$

PART-2 : CHEMISTRY

SECTION-I(i)

1. **Ans (A)**

$$\Delta S_{\text{Reaction}} = [186.264] - [5.740 + 2 \times 130.684]$$

$$\text{or } \Delta S_{\text{System}} = -80.844 \text{ J/K}$$

$$\Delta S_{\text{Surrounding}} = -\frac{\Delta H}{T} = \frac{74.81 \times 10^3}{298} = 251 \text{ J/K}$$

$$\Delta S_{\text{Universe}} = -80.844 + 251 = 170.2 \text{ J/K}$$

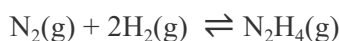
2. **Ans (C)**

$$N_{2(g)} = 7M = \text{Mass of } N_2$$

$$\text{Mole of } N_2 = \frac{M}{4}$$

$$H_{2(g)} = M = \text{Mass of } H_2$$

$$\text{Mole of } H_2 = \frac{M}{2}$$



$$\text{Pressure of } N_2 \propto \frac{M}{4}$$

$$\frac{P_0}{4} \quad \frac{P_0}{2}$$

$$\text{Pressure of } H_2 \propto \frac{M}{2}$$

$$\frac{P_0}{4} - x \quad \frac{P_0}{4} - 2x \quad x$$

$$\text{Assume Pressure of } N_2 = \frac{P_0}{4}$$

$$\frac{P_0}{4} + \frac{P_0}{2} - 2x = P = \text{Total pressure}$$

$$\text{Pressure of } H_2 = \frac{P_0}{2}$$

$$\frac{3P_0}{4} - 2x = P$$

$$x = \frac{P}{5}$$

$$P_0 = \frac{7P}{5} \times \frac{4}{3} = \frac{28P}{15}$$

$$\text{Partial pressure of } N_2$$

$$= \frac{P_0}{4} - x = \frac{7P}{15} - \frac{P}{5} = \frac{4P}{15}$$

$$\text{Partial pressure of } H_2$$

$$= \frac{P_0}{2} - 2x = \frac{14P}{15} - \frac{2P}{5} = \frac{8P}{15}$$

$$P_{H_2} = 2P_{N_2}$$

Only (C) option is correct.

PART-2 : CHEMISTRY

SECTION-I(ii)

7. **Ans (B,C)**

$$\frac{\ln K_1}{\ln K_2} > \frac{T_2}{T_1}$$

$$\Rightarrow T_1 \ln K_1 > T_2 \ln K_2$$

$$\Rightarrow -\Delta G^\circ_1 > -\Delta G^\circ_2$$

$$\Rightarrow (-\Delta H^\circ + T_1 \Delta S^\circ) > (-\Delta H^\circ + T_2 \Delta S^\circ)$$

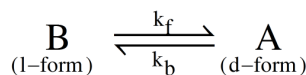
$$\Rightarrow T_1 \Delta S^\circ > T_2 \Delta S^\circ$$

$$\Rightarrow \Delta S^\circ > 0$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (7.00)**



$$t=0 \quad a \quad 0$$

$$t \quad a-x \quad x$$

$$t = t_{eq} \quad (a-x_{eq}) \quad x_{eq}$$

$$\text{at equilibrium } \frac{dx}{dt} = k_f(a-x_{eq}) - k_b x_{eq} = 0$$

$$\Rightarrow K_{eq} = \frac{k_f}{k_b} = \frac{x_{eq}}{a-x_{eq}}$$

$$\text{Where, } x_{eq} = \frac{a}{2} \Rightarrow k_f = k_b$$

$$(k_f + k_b) = \frac{2.303}{t} \log \left(\frac{x_{eq}}{x_{eq} - x} \right)$$

$$\Rightarrow 2k_f = \frac{2.303}{230.3} \log \frac{a/2}{a/2 - a/10} = \frac{1}{100} \log \frac{0.5}{0.4}$$

$$= \frac{1}{100} (0.7 - 0.6) = 10^{-3} \text{ s}^{-1}$$

$$\therefore k_f = 5 \times 10^{-4} \text{ s}^{-1} = k_b \quad \therefore k_{eq} = 1 \quad \Delta_r G^\circ = 0,$$

$$\Delta_r G = -ve$$

As A & B are enantiomers

$$\Rightarrow (S_m^\circ)_A = (S_m^\circ)_B$$

$$\text{and } \Delta_r H^\circ = 0, \Delta_r H = 0$$

2. **Ans (6.00)**

3d_{zx} orbital

(No radial node $\therefore n - \ell - 1 = 0$ & power of σ is

$$2 \Rightarrow \ell = 2)$$

$\therefore n = 3 \therefore 3d$. From $\Psi(\theta, \phi)$ orbital is d_{zx} , since

on changing to polar co-ordinate system, we get:

$$z = r \cos \theta; x = r \sin \theta \cos \phi; y = r \sin \theta \sin \phi$$

PART-3 : MATHEMATICS

SECTION-I(i)

1. **Ans (B)**

$$x^2 - x - 1 = 0 \quad \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$a\alpha^9 + b\alpha^8 + 1 = 0$$

$$a\beta^9 + b\beta^8 + 1 = 0$$

$$a(\alpha^8 - \beta^8) = -\frac{1}{\alpha} + \frac{1}{\beta}$$

$$a = 21, b = -34$$

2. **Ans (D)**

$$\text{Let } a + c + 2b = x \quad \dots\dots(1)$$

$$a + b + 2c = y \quad \dots\dots(2)$$

$$a + b + 3c = z \quad \dots\dots(3)$$

$$c = z - y; b = x + z - 2y$$

$$a = -x + 5y - 3z$$

$$\begin{aligned} & \left(\frac{2y-x}{x} \right) + \frac{4(x+z-2y)}{y} - \frac{8(z-y)}{z} \\ &= -17 + 2 \left(\frac{y}{x} + \frac{2x}{y} \right) + 4 \left(\frac{z}{y} + \frac{2y}{z} \right) \\ &= -17 + 4\sqrt{2} + 8\sqrt{2} \\ &= 12\sqrt{2} - 17 \end{aligned}$$

3. **Ans (D)**

$$\begin{aligned} f(n, r) &= \frac{1 \binom{n-1}{r-1} + 2 \binom{n-2}{r-1} + \dots + (n-r+1) \binom{r-1}{r-1}}{\binom{n}{r}} \\ &= \frac{\binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \dots + \binom{r}{r}}{\binom{n}{r}} \\ &= \frac{\binom{n+1}{r+1}}{\binom{n}{r}} = \frac{n+1}{r+1} \end{aligned}$$

4. **Ans (C)**

Number of ways to select

1 ice-cream = 1 (that is Vanilla)

2 ice-cream (VV, VB, VC, V K-P) = 4

3 ice-cream

one of them is Vanilla and remaining two can be selected as

(a) both diff. (b) both same

$${}^4C_2 = 6 \quad {}^5C_1 = 5$$

= 11 ways.

Now total ways (such that Vanilla is compulsory & Strawberry in pairs (i.e. 0,2,4,6)

$$= 4 \times (5+1) \times (4) \times (3+1) \times (2+1)$$

$$= 1152$$

required ways are

$$1152 - (1) - (4) - (11)$$

$$= 1136$$

5. **Ans (C)**

Since $x = -1$ and $x = 0$ are the two critical values of the absolute functions, one can suppose that

$$F(x) = a|x+1| + b|x| + cx + d$$

$$= \begin{cases} (c-a-b)x + d - a & \text{if } x < -1 \\ (a+c-b)x + a + d & \text{if } -1 \leq x \leq 0 \\ (a+b+c)x + a + d & \text{if } x > 0 \end{cases}$$

which implies that $a = 3/2$, $b = -5/2$, $c = -1$ and $d = 1/2$.

Hence

$$f(x) = (3x+3)/2, g(x) = 5x/2, \text{ and } h(x) = -x + \frac{1}{2}$$

6. **Ans (A)**

Applying c and d

$$\frac{\pi/2 + 2 \times \pi/2}{\sin^{-1} x^{2018} - \cos^{-1} x^{2018}} + 2 \{ \cos^{-1} x^{2017} - \sin^{-1} x^{2017} \} = \frac{-3}{-7}$$

$$\frac{3\pi/2}{4\cos^{-1} x^{2017}} - 2\cos^{-1} x^{2018} - \frac{\pi}{2} = \frac{3}{7}$$

$$\Rightarrow \frac{7\pi}{2} = 4\cos^{-1} x^{2017} - 2\cos^{-1} x^{2018} - \frac{\pi}{2}$$

$$2\cos^{-1} x^{2017} = 2\pi + \cos^{-1} x^{2018}$$

$$x = 1 \text{ \& } -1$$

$x = -1$ satisfies original equation

$$x = -1$$

$$\text{so, } \frac{\sin^{-1}(\sin \alpha) + \tan^{-1}(\tan \alpha)}{\cos^{-1}(\cos \alpha) + 2 \cos^{-1}(\cos \alpha)} = 2$$

PART-3 : MATHEMATICS

SECTION-I(ii)

7. Ans (A,C)

$$(A) {}^4C_1 \cdot {}^{3+2-1}C_{2-1} 2^{3 \cdot 2+2-1} C_{2-1}$$

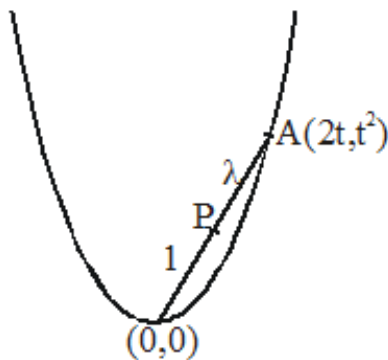
$$4.4.8.3 = 384$$

$$(B) {}^{3+3-1}C_{3-1} 3^{4 \cdot 3} = 15.243$$

$$(C) {}^{5+3-1}C_{3-1} {}^{3+3-1}C_{3-1} = \frac{6.7}{2} \cdot \frac{5.4}{2} = 210$$

$$(D) {}^{4+3-1}C_{3-1} {}^{2+3-1}C_{3-1} \{ {}^4C_1 2^3 + {}^4C_2 2^2 + {}^4C_3 2 + 1 \} \\ = \frac{6.5}{2} \cdot 6 \{ 32 + 24 + 8 + 1 \} = 5850$$

8. Ans (B,C,D)



$$\text{Let } \frac{a}{b} = \frac{1}{\lambda} (\lambda > 0)$$

$$P \left(\frac{2t}{1+\lambda}, \frac{t^2}{1+\lambda} \right)$$

$$\Rightarrow \frac{t^4}{(1+\lambda)^2} = \frac{2kt}{(1+\lambda)}$$

$$\Rightarrow \frac{t^3}{2k} = 1 + \lambda \Rightarrow \lambda = \frac{t^3 - 2k}{2k}$$

$$\Rightarrow \frac{t^3 - 2k}{2k} > 0 \Rightarrow \frac{2k - t^3}{2k} < 0 \Rightarrow k \in \left(0, \frac{t^3}{2} \right)$$

$$\because \beta < 16 \Rightarrow t^2 < 16 \Rightarrow t^3 < 64$$

$$k \in (0, 32) \Rightarrow \text{Sum of all integral values of } k$$

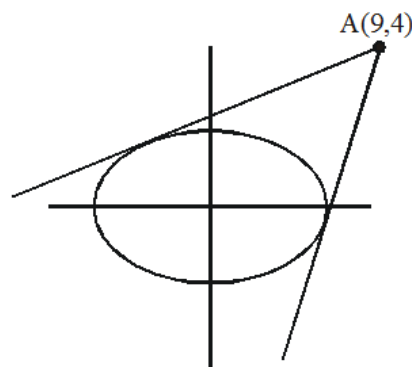
$$\text{is } \frac{32 \times 31}{2} = 496$$

$$\text{if } k = 4 \text{ then } \lambda = \frac{t^3 - 8}{8} \Rightarrow t^3 > 8 \Rightarrow t > 2$$

$$\Rightarrow 4 < \beta < 16$$

$$\text{if } \beta = 9 \Rightarrow t = 3 \quad \lambda = \frac{27 - 4}{4} = \frac{23}{4}$$

9. Ans (A,D)



$$\frac{y-4}{x-9} \text{ is the slope of line joining } A(9,4) \text{ and } (x,y)$$

For maximum and minimum value of this expression we have to determine the slope of tangents to the

$$\text{ellipse } \frac{x^2}{25} + \frac{y^2}{16} = 1 \text{ from } (9,4)$$

$$\text{Hence } y = Kx \pm \sqrt{16K^2 + 9}$$

it passes through (9,4)

$$\text{Hence } (4 - 9K)^2 = 16K^2 + 9 \Rightarrow 65K^2 - 72K + 7 = 0$$

$$\text{Hence } K = 1 \text{ or } \frac{7}{65} \Rightarrow M=1 \text{ \& } m = \frac{7}{65}$$

10. Ans (A,C)

$$f(x) = \begin{cases} \sqrt{9 - \log_2(x^2 + 4)} + \sqrt{\log_2(x^2 + 4) - 1}, & \text{if } x \in [0, 2) \\ \sqrt{9 - (\sqrt{9-x} + \sqrt{x-1})} + \sqrt{(\sqrt{9-x} + \sqrt{x-1}) - 1}, & \text{if } x \in [2, 4] \end{cases}$$

11. Ans (A,B,D)

$$f(x) = (2 - 2^{|x-1|}) \log_x (x^2 - 1)$$

Case-1 :

$$2 - 2^{|x-1|} \geq 0 \quad \log_x (x^2 - 1) \geq 0$$

$$\therefore x \leq 2 \quad x \geq \sqrt{2}$$

$$\Rightarrow \therefore x \in [\sqrt{2}, 2]$$

Case-2 :

$$2 - 2^{|x-1|} < 0 \quad \log_x (x^2 - 1) < 0$$

$$x > 2 \quad x \in (1, \sqrt{2})$$

Not possible.

$$\Rightarrow \text{Domain of } f(x) \text{ is } [\sqrt{2}, 2]$$

Hence option A,

$$\text{Since } f(x) = 0 \text{ at } x = \sqrt{2} \text{ and } 2$$

Hence option B is correct.

12. **Ans (C,D)**

Set of $[\sin^{-1}x] > [\cos^{-1}x]$ is $[\sin 1]$

Range of $|\sin x| + |\cos x|$ is $[1, \sqrt{2}]$

Range of $\left[x + \frac{1}{2}\right] + \left[x - \frac{1}{2}\right] + 2[-x]$ is $\{-1, -3\}$

Range of $\left[\frac{1}{[x-3]}\right]$ is $\{-1, 0, 1\}$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4.00)**

$$a_n = \frac{(2n+1)^9 C_{n+1}}{9} \text{ and } b_n = \frac{(3n+1)^8 C_{n+1}}{8}$$

a_n is max when $n+1=5$ and b_n is min when $n=0$

$\Rightarrow p=4$ and $q=0$

$p+q=4$

2. **Ans (6.00)**

Number divisible by 125 ends in 125, 250, 375, 625, 750, 875 (500 is not considered as 0 repeats)

4 digit number :

If it ends in 125, 375, 625, 875 then thousand's place can be filled in 6 ways using remaining non-zero digits

$\therefore$ Number of integers $= 4 \times 6 = 24$

If it ends in 250, 750 then thousands place can be filled in 7 ways

$\Rightarrow$ Number of integers $= 2 \times 7 = 14$

5 digit no. :

Case I : last three digit are 125, 375, 625 or 875

then remaining 2 places can be filled in 6×6 ways.

$\therefore$ such numbers are $6 \times 6 \times 4 = 144$

Case II : Last three digit are 250 or 750

remaining 2 places can be filled in 7×6 ways.

$\therefore$ such numbers are $7 \times 6 \times 2 = 84$

6 digit no. :

Case I : last three digit are 125, 375, 625 or 875

then remaining 3 places can be filled in $6 \times 6 \times 5$ ways.

$\therefore$ such numbers are $6 \times 6 \times 5 \times 4 = 720$

Case II : Last three digit are 250 or 750

remaining 3 places can be filled in $7 \times 6 \times 5$ ways.

$\therefore$ such numbers are $7 \times 6 \times 5 \times 2 = 420$

$\therefore k = \text{total numbers}$

$= 38 + 228 + 1140 = 1406 = 2 \times 19 \times 37$

Number of proper factors of $k = 2 \times 2 \times 2 - 2 = 6$

3. **Ans (4.00)**

Given,

$$(3x - y - 1) + b(4x - y - 2) = 0$$

$\therefore$ Fixed point is (1,2)

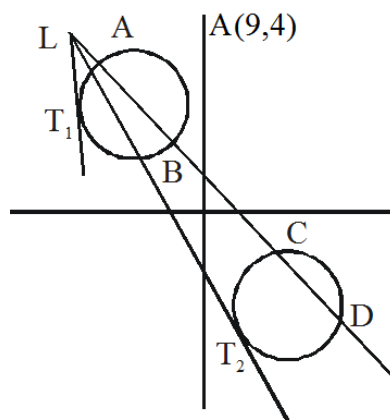
Let line be

$$(y - 2) = m(x - 1), m \in (-\infty, 0)$$

$$\therefore \text{Area}(m) = \frac{8}{2} + \frac{1}{2} \left(\sqrt{-m} - \frac{2}{\sqrt{-m}} \right)^2, m < 0$$

$$\Rightarrow A_{\text{Least}}(m = -2) = 4$$

4. **Ans (6.00)**



Locus of P will be circle in II & IV quadrant.

$$LA \cdot LB = LT_1^2$$

$$LC \cdot LD = LT_2^2$$

5. Ans (6.00)

$$\frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$a^2 = 16, b^2 = 1$$

$$a = 4, b = 1$$

let $A(x_1, y_1)$ and $B(x_2, y_2)$ then P, Q will be

mid point of AB will be $P \left(x_1, \frac{y_1}{4} \right)$ and

$$Q \left(x_2, \frac{y_2}{4} \right)$$

mid point of PQ will be

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{8} \right) = (h, k)$$

$\Rightarrow$ mid point of AB will be $(h, 4k)$

let point $S(t, t + 8)$ be a point on $y = x + 8$

AB is chord of contact of $x^2 + y^2 = 16$

$$\Rightarrow xt + y(t + 8) = 16 \quad \dots(i)$$

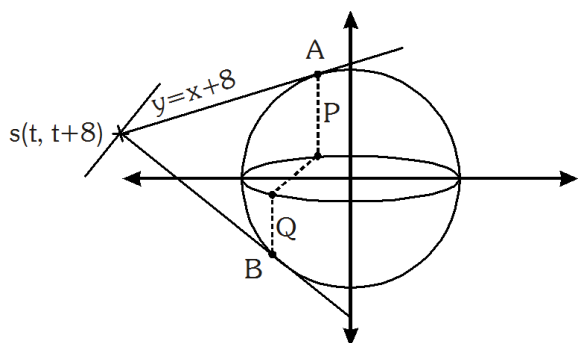
AB as chord whose mid point is $(h, 4k)$

$$xh + y(4k) = h^2 + 16k^2 \quad \dots(ii)$$

$$\frac{t}{h} = \frac{t + 8}{4k} = \frac{16}{h^2 + 16k^2}$$

$$\Rightarrow t = \frac{16h}{h^2 + 16k^2} \text{ and } t + 8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow \frac{16h}{h^2 + 16k^2} + 8 = \frac{64k}{h^2 + 16k^2}$$



$$\Rightarrow 2h + (h^2 + 16k^2) = 8k$$

$$\Rightarrow x^2 + 16y^2 + 2x - 8y = 0$$

$$x^2 + \alpha y^2 + \beta x + \gamma y = 0$$

$$\Rightarrow \alpha = 16, \beta = 2, \gamma = -8$$

$$\Rightarrow \alpha - \beta + \gamma = 16$$

6. Ans (21.00)

Coefficient of

$$x^{15} \rightarrow \left. \begin{array}{l} 1 x^5 x^{10} \rightarrow 1 \\ x x^4 x^{10} \rightarrow 5 \\ x^2 x^3 x^{10} \rightarrow 10 \\ x^2 x^5 x^8 \rightarrow 5 \end{array} \right\} 21$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,C,D	A,C,D	A,C,D	A,B,C,D	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.57 to 0.58	155.00	20.00	0.25	2.51	490.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	1	5	2	1	6

PART-2 : CHEMISTRY

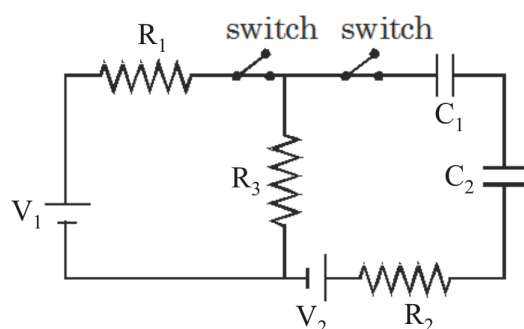
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	B,C,D	A,D	A,C	A,B,C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.00	1.00	6.00	4.00	2.50	1.41
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	8	2	5	7	4

PART-3 : MATHEMATICS

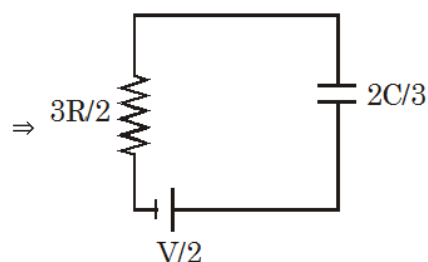
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C	A,B,C,D	A,C	A,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	221.00	12.50	22.00	2.50	13.00	16.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	7	5	4	7	3

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,B,D)



$$\frac{\frac{V_1}{R_1} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_3}} = \frac{V_1}{2} \Rightarrow$$



$$Q = \frac{C}{2} \times \frac{V}{2} (1 - e^{-\frac{1}{3}RC})$$

$$Q = \frac{CV}{4} (1 - e^{-1})$$

$$i = \frac{\frac{V}{2}}{\frac{3R}{2}} \times e^{-4/3} = \frac{V}{3R} e^{-4/3} ; \frac{V}{2} + \frac{iR}{2} = V_1 - i_1 R$$

$$i_1 = \frac{V}{2R} - \frac{i}{2} = \frac{V}{2R} (1 - 3e^{4/3}),$$

$$P_2 = V i = \frac{V^2}{3Re^{4/3}}$$

2. Ans (A,B,C,D)

According to gauss law

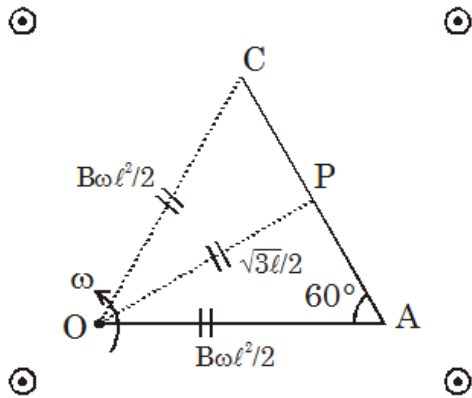
$$E 4\pi r^2 = \frac{q_0 - Q_t}{\epsilon_0}$$

$$J = \frac{E}{\rho} = \frac{q_0 - Q_t}{4\pi\epsilon_0\rho r^2}$$

$$I = J A = \frac{1}{\epsilon_0\rho} (q_0 - Q)$$

$$I = \frac{dQ}{dt} \Rightarrow \frac{dQ}{q_0 - Q} = \frac{dt}{\epsilon_0\rho} \Rightarrow \frac{Q}{q_0} = 1 - e^{-t/\epsilon_0\rho}$$

4. Ans (A,C,D)



$$\epsilon_{AC} = 0$$

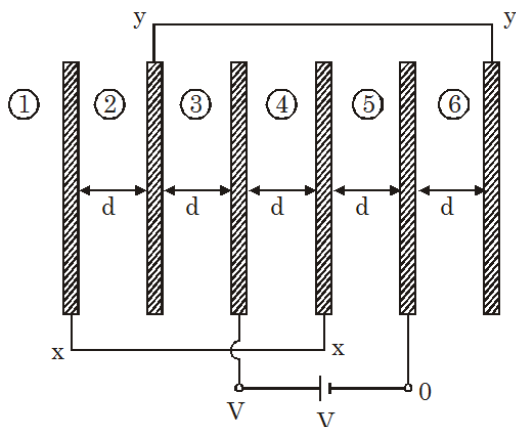
$$\epsilon_{OP} = \frac{B\omega 3l^2}{8}$$

$$V_C - V_P = \frac{B\omega l^2}{2} - \frac{3B\omega l^2}{8}$$

$$= \frac{B\omega l^2}{2}$$

$$\text{similarly } V_A - V_P$$

5. Ans (A,B,C,D)



$$C(y - x) + C(y - V) + C(y - 0) = 0$$

$$3y - x = V$$

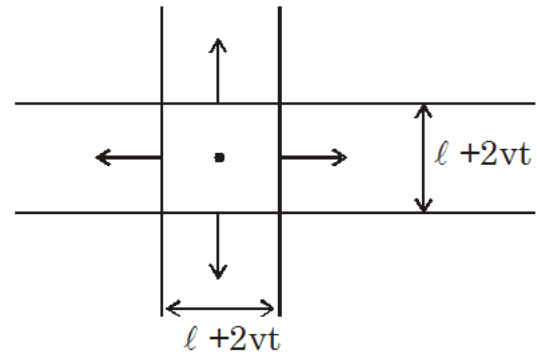
$$C(x - y) + C(x - V) + C(x - 0) = 0$$

$$3 \times 3x - y = V$$

$$3y - x = V$$

$$x = \frac{V}{2}, y = \frac{V}{2}$$

6. Ans (A,D)



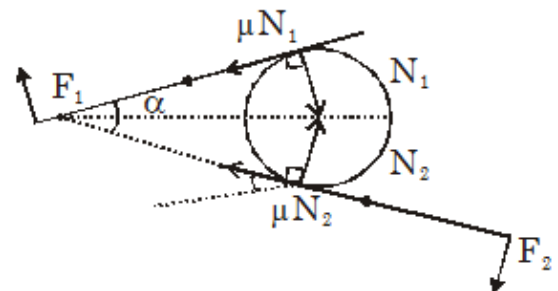
$$e = B(\ell + 2vt)v \times 4$$

$$i = \frac{\epsilon}{r} = \frac{4B(\ell + 2vt)}{P(\ell + 2vt)}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.57 to 0.58)



$$N_1 = N_2$$

$$2\mu N \cos 30^\circ = 2N \cos 60^\circ$$

$$\Rightarrow \mu = \frac{\cos 60^\circ}{\cos 30^\circ} = \frac{1}{\sqrt{3}}$$

2. Ans (155.00)

$$i = 5 + 2t$$

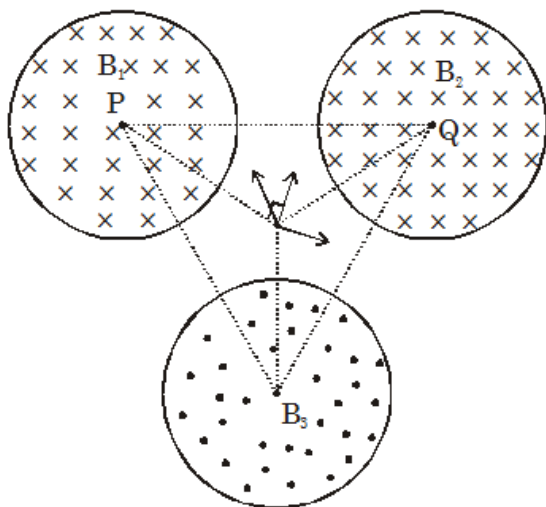
$$H = \int i^2 R dt = 60 \int_0^{10} (5 + 2t)^2 dt$$

$$= 60 \left[25t + \frac{4t^3}{3} + \frac{20t^2}{2} \right]$$

$$= 60 \left[25 \times 10 + \frac{4}{3} \times 1000 + 10 \times 100 \right]$$

$$= 60 \left[\frac{7750}{3} \right] = 155 \text{ kJ}$$

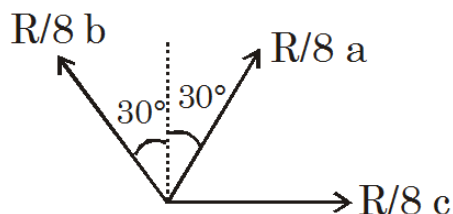
4. Ans (0.25)



$$E = \frac{R^2}{2x} \frac{dB}{dt} = \frac{R^2}{2} \times \frac{dB}{dt}$$

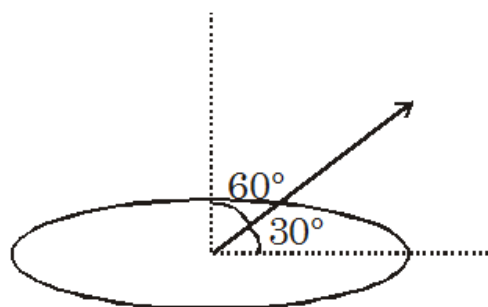
$$= \frac{R^2}{2 \times \frac{2}{3} \times \frac{\sqrt{3}}{2} 4\sqrt{3}} \times \frac{dB}{dt}$$

$$= \frac{R}{8} \frac{dB}{dt}$$



$$E = \frac{R}{8} \sqrt{(\sqrt{3})^2 + 1^2} = 0.25 \text{ V/m}$$

5. Ans (2.51)



$$Q = \frac{\Delta\phi}{R} = \frac{BA \cos 60^\circ}{R}$$

$$= \frac{10^{-2} \times \pi \times 16 \times 10^{-4}}{0.01} \times \frac{1}{2}$$

$$= 8\pi \times 10^{-4} \text{ C}$$

$$= 2.51 \times 10^{-3} \text{ C}$$

6. Ans (490.00)

$$\omega = \alpha t = 7.5 \text{ rad/s}$$

$$73.5 = I\omega = I \times 7.5$$

$$I = \frac{735}{75}$$

$$k = \frac{1}{2} I \omega^2 = \frac{1}{2} \times \frac{49}{5} \times 10^2 = 490 \text{ J}$$

PART-1 : PHYSICS

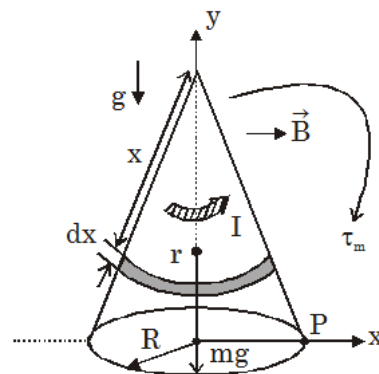
SECTION-III

1. Ans (2)

$$\text{magnetic torque } \vec{\tau}_m = \vec{\mu} \times \vec{B}$$

for no toppling $\tau_{mg} > \tau_m$

The cone can be assumed as the collection of many current carrying rings. Each ring will have a dipole moment associated with it.



Rough Surface

$$dI = \frac{I}{\pi RL} 2\pi r \cdot dx = \frac{2Irdx}{RL}$$

$$du = dI\pi r^2$$

$$= \frac{2dI\pi r^3 dx}{RL}$$

$$\frac{r}{R} = \frac{x}{L}$$

$$du = \frac{2I\pi R^3 x^3 dx}{RL^4}$$

$$\int du = \frac{2I\pi R^3}{L^4} \int_0^L x^3 dx$$

$$u = \frac{xIR^2}{2} \uparrow$$

for no toppling

$$\tau_{mg} > \tau_m$$

$$mgR > \frac{\pi IR^2 B}{2}$$

$$I < \frac{2mg}{\pi BR} \quad I_{\max} = \frac{Kmg}{\pi BR}$$

$$\Rightarrow K = 2$$

2. Ans (1)

$$W_B = W_{\text{ext}} - \Delta U = 0$$

$$W_{\text{ext}} = \Delta U - W_B$$

$$Q_i = KCV$$

$$Q_f = CV$$

$$\Delta Q = (1 - K)CV$$

$$W_B = (1 - K)CV^2$$

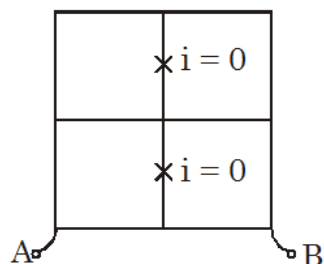
$$W_{\text{ext}} = \left(\frac{1}{2} CV^2 - \frac{1}{2} KCV^2 \right) + (K - 1)CV^2$$

$$= (K - 1) \frac{CV^2}{2}$$

$$= \frac{4 \times \epsilon_0 \times 200\pi \times 10^{-4} \times 300}{2 \times 10^{-4}}$$

$$= \frac{(200) \times 10^{-4} \times 300}{2 \times 10^{-3} \times 9 \times 10}$$

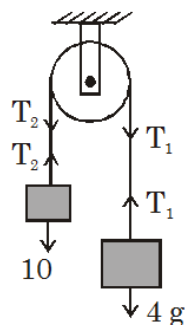
3. Ans (5)



$$\Rightarrow R_{\text{eq}} = \frac{\left(\frac{16 \times 8}{16+8} + 8 \right) \times 8}{8 + \frac{16 \times 8}{16+8} + 8}$$

$$= \frac{\frac{40}{3} \times 8}{16 + \frac{16}{3}} = \frac{40 \times 8}{64} = 5\Omega$$

4. Ans (2)



$$40 - T_1 = 4 \times 5$$

$$T_1 = 20\text{N}$$

$$T_2 - 10 = 1 \times 5$$

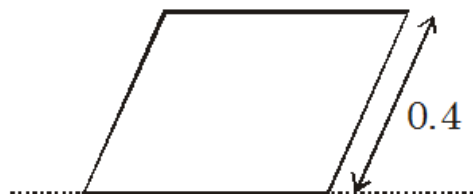
$$T_2 = 15\text{N}$$

$$T_1 r - T_2 r = I \frac{a}{r}$$

$$\frac{5r^2}{5} = \frac{1}{2} mr^2 \Rightarrow 2\text{ kg}$$

5. Ans (1)

$$I = m \frac{(0.4)^2}{3}$$



$$I' = m \frac{(0.2)^2}{3}$$



$$\omega = \alpha t$$

$$= \frac{\tau}{I} t$$

$$t = \frac{I\omega}{\tau}$$

$$t' = \frac{I'\omega}{\tau'}$$

$$\frac{t'}{t} = \frac{I'}{I} = \frac{1}{4}$$

$$t' = 2\text{ sec}$$

6. Ans (6)

$$\frac{d\phi}{dt} = 10^{-2} \times -2B_0 C t$$

$$i = \frac{10^{-2} \times -2B_0 C t}{R}$$

$$t_{\text{max}} = \sqrt{\frac{a}{c}} = 3$$

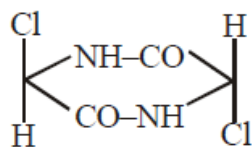
$$i = \frac{10^{-2} \times -2 \times 0.01 \times 1 \times 3}{10^{-2}}$$

$$= 6 \times 10^{-3} \text{ A}$$

PART-2 : CHEMISTRY

SECTION-I

4. Ans (A,C)



- (i) It has centre of symmetry
- (ii) It has no plane of symmetry
- (iii) It is optically inactive
- (iv) It can not rotate PPL

Therefore only (A) & (C) are correct

PART-2 : CHEMISTRY

SECTION-II

1. Ans (5.00)

$$W_{xy_3} = 1 \text{ gm}$$

$$M_{xy_3} = x + 3y$$

$$\Delta T_f = 1^\circ\text{C}$$

$$k_f = 5 \text{ K molal}^{-1}$$

$$W_{\text{solvent}} = 100\text{g} = 0.1 \text{ kg}$$

$$\Rightarrow \Delta T_f = mK_f$$

$$\Rightarrow \Delta T_f = \frac{\frac{1}{x+3y} \times 5}{0.1}$$

$$\Rightarrow 1 = \frac{1}{0.1(x+3y)} \times 5$$

$$\Rightarrow x + 3y = 50 \dots (1)$$

$$W_{xy} = 0.5\text{g}$$

$$M_{xy} = x + y$$

$$n_{xy} = \frac{0.5}{x+y}$$

$$\Delta T_f = 1^\circ\text{C}$$

$$K_f = 5 \text{ k molal}^{-1}$$

$$W_{\text{solvent}} = 100\text{g} = 0.1 \text{ kg}$$

$$\Rightarrow \Delta T_f = mK_f$$

$$\Rightarrow \Delta T_f = \frac{n_{xy}}{W_{\text{solvent}}} \times k_f$$

$$\Rightarrow 1 = \frac{\frac{0.5}{x+y} \times 5}{0.1}$$

$$\Rightarrow x + y = 25 \dots (2)$$

2. Ans (1.00)

With NaOH valency factor of $K_xH_y(C_2O_4)_z \cdot nH_2O$ is y

Mili equivalents of salt = mili equivalents of NaOH

$$\frac{2.54}{m} \times 10 \times y = 0.01 \times 30 \dots (1)$$

M = molecular weight of salt.

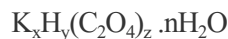
With $KMnO_4$ valency factor of $K_xH_y(C_2O_4)_z \cdot nH_2O$ is $2z$

$$\frac{2.54}{m} \times 10 \times 2z = 0.01 \times 40 \dots (2)$$

$$\frac{\text{eq. 1}}{\text{eq. 2}} = \frac{y}{2z} = \frac{3}{4}$$

$$\frac{y}{z} = \frac{3}{2}$$

Apply charge balancing



$$x + y - 2z = 0 \quad (y = 3, z = 2)$$

$$x + 3 - 4 = 0$$

$$x = 1$$



Calculate 'm' from eq. 1

$$m = 254$$

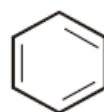
$$39 + 3 + 88 \times 2 + 18n = 254$$

$$n = 2$$

Hence $x = 1, y = 3, z = 2$

$$\text{Ans. } \frac{1 + 3 - 2}{2} = 1$$

3. Ans (6.00)



(Benzene)

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (8)**

$$U_{\text{rms}} = \sqrt{\frac{3RT}{M}} \propto \sqrt{T}$$

$$T \propto U_{\text{rms}}^2$$

$$\frac{T_2}{T_1} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

In reversible adiabatic process

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \Rightarrow \frac{V_2}{V_1} \left(\frac{T_1}{T_2}\right)^{1/\gamma-1}$$

$$\frac{V_2}{V_1} = (4)^{3/2} = 8$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B,C)**

L'S - 2

C, O, G → 3 different letters

E'S - 2

(1) Four different letters = ${}^5C_4 = 5$

(2) 2 like 2 different ${}^2C_1 \times {}^4C_2 = 12$

(3) 2 alike, 2 alike = 1 = 1

18

2. **Ans (A,B,C)**

According to the given conditions

$a > 0, b^2 < 4ac; b > 0, c^2 < 4ab, c > 0, a^2 < 4bc$

$\therefore a^2 + b^2 + c^2 < 4(bc + ca + ab)$

$$\Rightarrow \frac{1}{4} < \alpha$$

Also $a^2 + b^2 + c^2 - (bc + ca + ab)$

$$= \frac{1}{2} [(b-c)^2 + (c-a)^2 + (a-b)^2] > 0$$

$$\Rightarrow \alpha < 1$$

3. **Ans (A,B,C,D)**

$$0 < \cot^{-1}x < \pi$$

$$\text{sgn}(\cot^{-1}x) = 1$$

$$2n < x < 2n + 1 \Rightarrow f(x) = 1$$

$$2n + 1 < x < 2n + 2 \Rightarrow f(x) = \infty \text{ (undefined)}$$

so many one and periodic and bound always +ve.

4. **Ans (A,C)**

$\text{gof}(x)$ exists when $f(x) \geq 1$ and $x \in [-1, 4]$

$$x^2 \geq 1 \Rightarrow x \in (-\infty, -1] \cup [1, \infty)$$

$$\Rightarrow x \in [1, 4] \cup \{-1\}$$

$$\text{Range of } \text{gof}(x) = [0, \sec^{-1}16]$$

5. **Ans (A,D)**

$$\therefore 81^n = (1 + 80)^n = 1 + 80\lambda, \lambda \in \mathbb{N}$$

$$\therefore 3^{81^n} = 3^{1+80\lambda} = 3 \cdot 3^{80\lambda} = 3 \cdot (9)^{40\lambda} = 3(10-1)^{40\lambda}$$

$$= 3(1 + 10\mu) = 3 + 30\mu$$

$$\therefore \text{Last digit of } 3^{3^{4n}} + 1 \text{ is } 4.$$

6. **Ans (A,B)**

Each selection of 4 points, two on one line and two on other will give one point of intersection.

$$\therefore \text{Required number of intersection} = {}^mC_2 \cdot {}^nC_2$$

$$= \frac{m(m-1)}{2} \cdot \frac{n(n-1)}{2} = \frac{mn(m-1)(n-1)}{4}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (221.00)**

$$b = -\sum_{r=1}^{50} a_r$$

$$\text{But } a_r = r^2 \frac{(50-r+1)}{r} = \sum r(51-r)$$

2. Ans (12.50)

$$x^2 + y^2 - 4x(\ell x + my) + 3(\ell x + my)^2 = 0$$

coefficient of x^2 + coefficient of y^2 = 0

$$1 - 4\ell + 3\ell^2 + 1 + 3m^2 = 0$$

$$2 - 4\ell + 3(\ell^2 + m^2) = 0 \quad \dots(i)$$

Let foot of $\perp$ from origin on

$\ell x + my - 1 = 0$ is (x, y)

$$\frac{x-0}{\ell} = \frac{y-0}{m} = -\left(\frac{-1}{\ell^2 + m^2}\right)$$

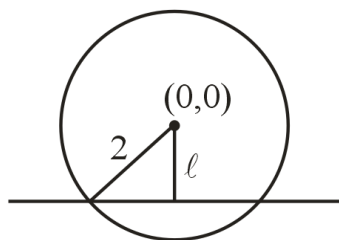
$$x = \left(\frac{\ell}{\ell^2 + m^2}\right) \quad y = \left(\frac{m}{\ell^2 + m^2}\right) \quad \dots(ii)$$

$$x^2 + y^2 = \frac{1}{\ell^2 + m^2} \quad \dots(iii)$$

after putting values from equation (ii) and (iii)

in equation (i) we get $\lambda = 2$

3. Ans (22.00)



$$\text{Length of chord} = 2\sqrt{4 - \ell^2}$$

$$\text{where } \ell = \left| \frac{n}{\sqrt{2}} \right|$$

$$= 2\sqrt{4 - \frac{n^2}{2}} = \sqrt{2}\sqrt{8 - n^2}$$

$$2(7) + 2(4) = 22$$

4. Ans (2.50)

$$\begin{aligned} S_n &= \frac{1}{2} \sum_{r=1}^n \frac{(2r+1) - 1}{1.3.5 \dots (2r-1)(2r+1)} \\ &= \frac{1}{2} \sum_{r=1}^n \left[\frac{1}{1.3.5 \dots (2r-1)} - \frac{1}{1.3.5 \dots (2r+1)(2r+1)} \right] \\ &= \frac{1}{2} \left[\frac{1}{1} - \frac{1}{1.3} \right] + \frac{1}{2} \left[\frac{1}{1.3} - \frac{1}{1.3.5} \right] \\ &\quad + \frac{1}{2} \left[\frac{1}{1.3.5} - \frac{1}{1.3.5.7} \right] + \dots + \\ &\quad + \frac{1}{2} \left[\frac{1}{1.3.5 \dots (2n-1)} - \frac{1}{1.3.5.7 \dots (2n+1)} \right] \\ S_n &= \frac{1}{2} \left[1 - \frac{1}{1.3.5.7 \dots (2n+1)} \right] \\ S_\infty &= \frac{1}{2} \end{aligned}$$

$$5S_\infty = 2.5$$

5. Ans (13.00)

Let $\log_2 x = t$

$$t^4 - (3t-3)^2 + 9(5-2t) < 4t^2$$

$$(t^4 - 4)(t^2 - 9) < 0$$

$$t \in (-3, -2) \cup (2, 3)$$

$$\therefore x \in \left(\frac{1}{8}, \frac{1}{4} \right) \cup (4, 8)$$

6. Ans (16.00)

$$\therefore S = \frac{\cos \frac{2\pi}{65} + \cos \frac{6\pi}{65} + \dots + \cos \frac{62\pi}{65}}{\cos \frac{2\pi}{65} \cdot \cos \frac{4\pi}{65} \dots \cos \frac{32\pi}{65}} = 16$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (4)

$$x + y + z = 10$$

$$1 \quad 2 \quad 7$$

$$3 \quad 1 \quad 6$$

$$4 \quad 1 \quad 5$$

$$2 \quad 3 \quad 5$$

Total ways = 4.

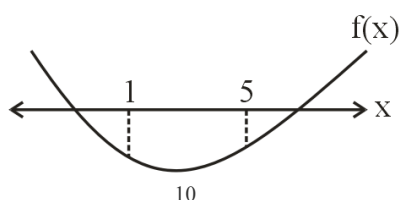
2. Ans (7)

$$\frac{(a+2) + (a+2) + b + c}{4} \geq \left((a+2)^2 \cdot bc \right)^{1/4} \geq (81)^{1/4}$$

$$a+2 = b = c = 3$$

$$a = 1, b = c = 3$$

3. Ans (5)



We have $1 + \sum_{r=1}^{10} (3^r \cdot {}^{10}C_r + r \cdot {}^{10}C_r)$

$$= 1 + \sum_{r=1}^{10} 3^r \cdot {}^{10}C_r + 10 \sum_{r=1}^{10} {}^9C_{r-1}$$

$$= 1 + 4^{10} - 1 + 10 \cdot 2^9$$

$$= 4^{10} + 5 \cdot 2^{10} = 2^{10} (4^5 + 5) = 2^{10} (\alpha \cdot 4^5 + \beta), \text{ so } \alpha = 1$$

and $\beta = 5$

Now $f(1) < 0$ and $f(5) < 0$

So, $f(1) < 0 \Rightarrow -k^2 < 0$

$$\Rightarrow k \neq 0$$

and $f(5) < 0 \Rightarrow 16 - k^2 < 0$

$$\Rightarrow k^2 - 16 > 0$$

$$k \in (-\infty, 4) \cup (4, \infty)$$

Hence smallest positive integral value of $k = 5$

4. Ans (4)

We must have $x \geq 1$ and $y \geq 2$, since x is less than y . That is, $y = j$ with $2 \leq j \leq 9$, then x can take the $j - 1$ values from 1 to $j - 1$ and z can take the j values from 0 to $j - 1$. For each value $y = j$, therefore, xyz can take $j(j - 1)$ values. We can now get the answer by summing over all values of y :

$$\sum_{j=2}^9 j(j-1) = \sum_{j=1}^9 j(j-1) = \sum_{j=1}^9 j^2 - \sum_{j=1}^9 j$$

$$= 285 - 45 = 240$$

5. Ans (7)

$$\cos\left(\frac{1}{4} \tan^{-1} \frac{24}{7}\right) = \cos\left(\frac{1}{4} \cos^{-1} \frac{7}{25}\right)$$

$$= \cos\left(\frac{1}{4} \theta\right) \text{ where } \theta = \cos^{-1} \frac{7}{25}$$

$$\cos \theta = \frac{7}{25}$$

$$2 \cos^2 \frac{\theta}{2} = 1 + \cos \theta$$

$$2 \cos^2 \frac{\theta}{2} = \frac{32}{25}$$

$$\cos^2 \frac{\theta}{2} = \frac{16}{25}$$

$$\cos \frac{\theta}{2} = \frac{4}{5}$$

$$2 \cos^2 \frac{\theta}{4} = 1 + \cos \frac{\theta}{2} = 1 + \frac{4}{5}$$

$$2 \cos^2 \frac{\theta}{4} = \frac{9}{5}$$

$$\cos^2 \frac{\theta}{4} = \frac{9}{10}$$

$$\cos \frac{\theta}{4} = \frac{3}{\sqrt{10}}$$

6. Ans (3)

$$\frac{x^3 + 1}{2} = 2 \sqrt[3]{2x - 1}$$

Let $f(x) = \frac{x^3 + 1}{2} \Rightarrow f^{-1}(x) = \sqrt[3]{2x - 1}$

Equation becomes $f(x) = f^{-1}(x)$

$$\Rightarrow f(x) = x \Rightarrow \frac{x^3 + 1}{2} = x \Rightarrow x^3 - 2x + 1 = 0$$

$$\Rightarrow (x - 1)(x^2 + x - 1) = 0 \Rightarrow x = 1, \frac{-1 \pm \sqrt{5}}{2}$$